Leong Ko Ryan Jasper

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Singapore Citizen — eligible for H-1B1 (US–Singapore FTA), a no-lottery work visa similar to TN

Computer Engineering undergraduate with hands-on experience in robotics software, embedded systems, visual–inertial SLAM, and FPGA-based timing systems.

Education

National University of Singapore Aug 2023 – Aug 2026
Bachelor of Engineering (Honors), Major in Computer Engineering *Singapore*

- GPA: 4.88/5.0; Dean's List; Engineering Scholars Program

Stanford University Apr 2025 – Jun 2025
Visiting Student, Spring Quarter (Exchange) *Stanford, CA*

Experience

Embedded / Robotics SWE Jan 2025 – Present
ZeroshotData *San Francisco, CA*

- Designed and assembled PCBs for EMF tracking emitter/receiver system using KiCad, programming STM32 microcontrollers; designed Helmholtz cage in OnShape/CAD with SCPI automation on OrangePi
- Set up Piper 6 DOF robotic arms and Raspberry Pi Zeros for automated training data collection and puppet gripper control, enabling large-scale dataset generation for deep learning models
- Integrated OAK-D-WIDE stereo cameras with DepthAI C++ bindings, implementing timestamp drift correction and container lifecycle management for high-bitrate H.264 streams
- Built battery monitoring service in C++ with I^2C debugging and MQTT publishing; developed OTA firmware update framework for Wi-Fi-based microcontroller updates

Embedded Engineer Intern May 2024 – Aug 2024
*A*STAR — National Metrology Centre* *Singapore*

- Engineered frequency synthesizer using Verilog and GOWIN IP cores on Tang Nano 20K, achieving 6.5–9.0 ns precision; developed autonomous data-measurement web server using Python on OrangePi 3B+

Projects

Remote-controlled Robot with SLAM | *Bare-metal Programming, C++, ROS2* Mar 2024 – Apr 2024

- Designed ROS 2-based robotic system integrating LiDAR, motor control, and real-time mapping with sensor fusion for autonomous navigation

Flair AI | *AI writing assistant; React, Tiptap, Twilio, Brainbase, Gemini, Tailwind* 2025

- **Winner (1st place, $1,500)** — AI for Good Hackathon, San Francisco. Built end-to-end application with automated grading and AI voice agent for ~400 students; demo video

Skills

Embedded & Hardware: Verilog/SystemVerilog, FPGA (Basys-3, Tang Nano), Microcontrollers (STM32, ESP32, Arduino, Raspberry Pi, Raspberry Pi Zero, OrangePi), FreeRTOS, I^2C, SPI, CAN, UVC, RTL design, KiCad, PCB design
Robotics & Systems: C/C++, ROS 2, SLAM, LiDAR, Linux, Docker, Git, Real-time systems, MQTT, DepthAI, SCPI, 6 DOF arms
CAD & Simulation: OnShape, CAD, Electromagnetic simulation
Computer Vision & Cloud: OpenCV, FFmpeg, Python, Kalibr, GCP, Kubernetes, Terraform
Portfolio: https://portfolio-version2-five.vercel.app/